

90-GHz Cascaded Microring Modulators Enabling 4.8-Tb/s Optical Interconnects at 600 Gb/s/λ

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Abstract We demonstrate an 8-channel O-band cascaded MRM transmitter featuring ~90-GHz 3-dB electro-optic bandwidth without electrical peaking. The optimized carrier-depletion junction and low-parasitic electrodes enable a broadband electro-optic response for 200-Gbaud PAM8, achieving a record 600 Gb/s/λ and 4.8-Tb/s aggregate capacity. ©2026

Introduction

The rapid scaling of large language models (LLMs), particularly mixture-of-experts (MoE) architectures, is driving unprecedented communication demands in large-scale AI clusters [1, 2]. In MoE systems, expert parallelism requires frequent all-to-all data exchanges among thousands of accelerators, making both high aggregate bandwidth and low latency critical for kilometre-scale interconnects.

Optical interconnects based on wavelength-division multiplexing (WDM) and silicon microring modulators (MRMs) offer high-capacity and high-density solutions for kilometre-scale AI interconnects [3–9]. Multi-channel MRMs have recently advanced in both wavelength parallelism and per-wavelength data rate, as summarized in Fig. 1. A 5-channel silicon MRM array achieved 200 Gb/s/λ and 1 Tb/s aggregate capacity [3], while an 8-channel DWDM transmitter reached 256 Gb/s/λ and 2.05 Tb/s [4]. More recently, 448-Gb/s transmission at 224 Gbaud using 4-level pulse amplitude modulation (PAM4) has been demonstrated with silicon MRM technology [9]. Hollow-core fiber (HCF) enables lower-latency transmission over kilometre-scale links, with a recent 448-Gb/s demonstration over 5.1 km achieving a 28.4% latency reduction [10]. However, previous demonstrations have focused on either high per-wavelength data rates, multi-channel capacity, or low-latency transmission.

In this work, we demonstrate an 8-channel cascaded O-band MRM transmitter featuring a measured ~90-GHz 3-dB electro-optic bandwidth without electrical peaking. The optimized carrier-depletion PN junctions and low-parasitic electrodes enable a broadband electro-optic response for high-speed multilevel modulation. Operating at up to 200-Gbaud PAM8, the transmitter achieves 600 Gb/s/λ and 4.8-Tb/s aggregate raw capacity. Clear PAM8 eyes and bit error ratios

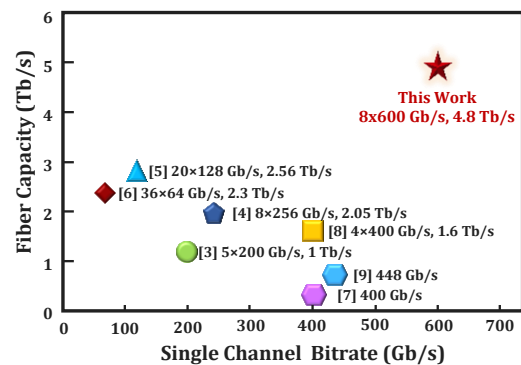


Fig. 1: Recent demonstrations on high-speed and multi-channel MRMs.

(BERs) of approximately $5e-2$ are achieved across all channels after 2-km transmission over a single HCF.

8-Channel Microring Modulators

The 8-channel cascaded MRM array was fabricated on a 12-inch commercial silicon-on-insulator (SOI) platform. Figures 2(a) and (b) show the three-dimensional schematic and cross-sectional views of the device, respectively. The device consists of a 220-nm-thick silicon rib waveguide with a 70-nm slab on a 2-μm buried-oxide layer. Each microring is designed in an add-drop configuration with a compact radius of approximately 4.0 μm, yielding a measured free spectral range (FSR) of about 17 nm in the O-band [11,12]. To simultaneously achieve low propagation loss and single-mode operation, the bus waveguide width is set to 400 nm, while the ring waveguide is widened to 600 nm to suppress bending loss [13,14]. A bent directional coupler with a 150-nm coupling gap at the through port and a 350-nm coupling gap at the drop port is utilized to achieve near-critical coupling and a high extinction ratio.

The carrier-depletion PN junction is optimized for high-speed operation. An asymmetric lateral

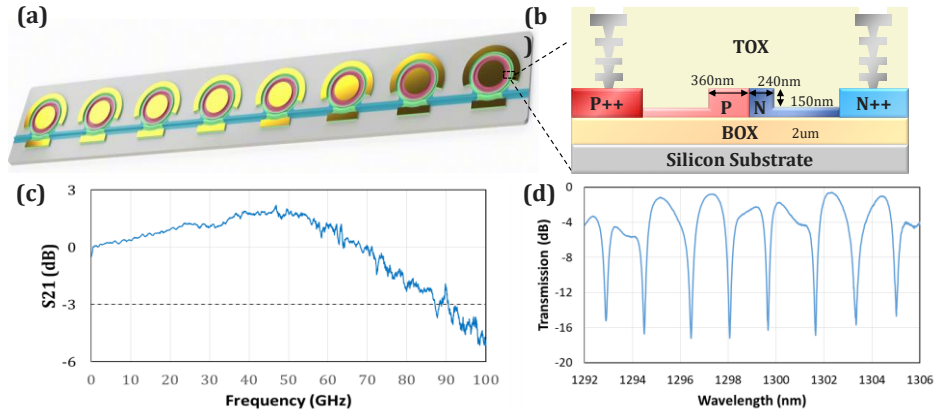


Fig. 2: (a) 3D schematic, (b) cross-sectional view, (c) EO S21, and (d) optical spectra of the 8-MRM, respectively.

junction with 360-nm P and 240-nm N regions compensates for the optical-mode shift toward the outer side of the bend, thereby improving electro-optic overlap and modulation efficiency. Multiple implantation levels define the depletion region, while highly doped P⁺/N⁺ and P⁺⁺/N⁺⁺ regions and Salicide contacts reduce series resistance and parasitic electrical losses.

Eight MRMs are cascaded along a common bus waveguide, with intentionally staggered resonance wavelengths to enable about 300GHz-spaced WDM. Eight independent ground-signal-ground (GSG) radio frequency (RF) pad sets with a 125- μ m pitch provide high-speed electrical access while minimizing inter-channel RF crosstalk. Heater is used to tune the operating wavelength of each MRM. Moreover, edge coupling is employed at the chip facets to provide wavelength-insensitive coupling loss.

Benefiting from the optimized PN junction and low-parasitic electrode design, the typical measured EO response of the MRMs exhibits a 3-dB bandwidth of about 90 GHz at 3-dB insertion-loss detuning, followed by a gradual roll-off extending beyond 100 GHz, as shown in Fig. 2(c). The optical spectra in Fig. 2(d) exhibit a loaded Q-factor of approximately 3600 and a high extinction ratio of $\sim$ 17 dB, with well-resolved resonance peaks across all eight channels.

Experimental setup

Figure 3 illustrates the experimental setup. The output of a tunable continuous-wave (CW) polarization-controlled laser is launched into the 8-channel cascaded MRM with 14dBm optical power. Electrical PAM4 and PAM8 signals at various baud rates are generated by a 256-GSa/s digital-to-analog converter (DAC) and amplified by 110-GHz RF drivers to approximately 1 Vpp. High-speed electrical probes deliver the amplified signals to the corresponding MRMs.

The reverse bias and optical carrier detuning of the MRMs are carefully adjusted to optimize the modulation response. This optimization balances modulation efficiency, electro-optic bandwidth, and amplitude linearity for multilevel PAM signals. For 200-Gbaud PAM8 modulation, each wavelength carries 600 Gb/s, resulting in an aggregate raw line rate of 4.8 Tb/s.

The optical output is coupled through a 2-km HCF. The employed HCF exhibits a measured transmission loss of 0.23 dB/km and chromatic dispersion of $\sim$ 1 ps/(nm $\cdot$ km) in the operating wavelength range. The propagation delay of the 2-km HCF is $\sim$ 6.7 μ s, corresponding to approximately 30% lower latency than conventional SSMF over the same distance.

At the receiver side, a Pr-doped fiber amplifier

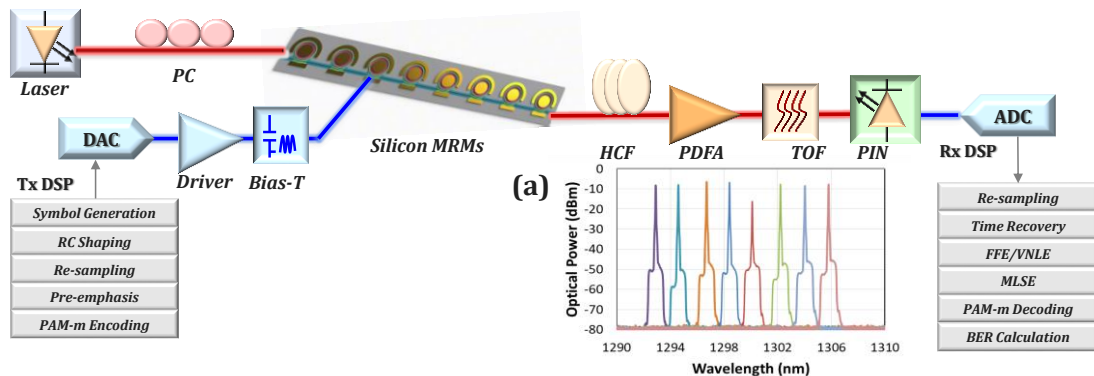


Fig. 3: Experimental setup for 8-MRM transmission over 2-km HCF. Inset (a) optical spectra of 8-channel signals.

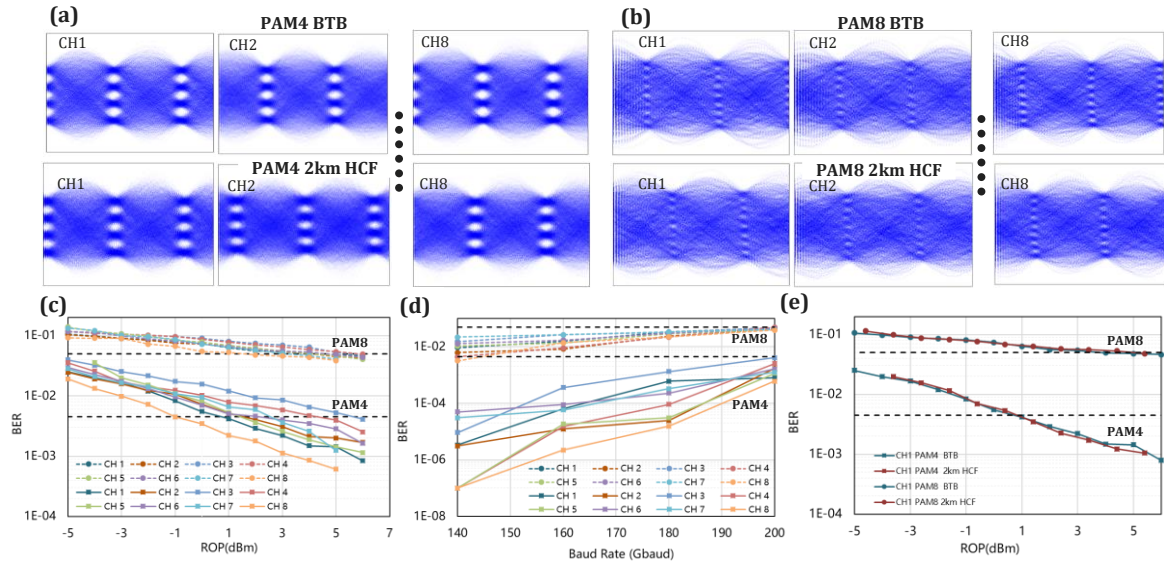


Fig. 4: (a, b) Eye diagrams of 200-Gbaud PAM4 and PAM8 signals for all 8 channels before and after 2-km HCF transmission. (c) BTB BER performance of the 8 channels for 200-Gbaud PAM4 and PAM8. (d) BTB BER performance of PAM4 and PAM8 signals at different baud rates. (e) Transmission penalties of 200-Gbaud PAM4 and PAM8 after 2-km HCF.

(PDF) with 18-dB gain is used to compensate for link loss. A 1.6-nm tunable optical filter suppresses amplified spontaneous emission (ASE) noise before high-speed detection. The received optical signal is captured by a high-bandwidth analog-to-digital converter (ADC) for offline processing.

The transmitter digital signal processing (DSP) consists of PAM symbol generation, raised-cosine pulse shaping, nonlinear pre-distortion, and pre-emphasis to compensate for the frequency responses of the electrical and optical components. At the receiver, the signal is processed offline using resampling, timing recovery, feed-forward equalization (FFE), a third-order Volterra nonlinear equalizer (VNLE), maximum-likelihood sequence estimation (MLSE), and PAM-m decoding. The recovered bits are then compared with the transmitted bits to calculate the BER. The optical spectra of the eight modulated channels are shown in the inset of Fig. 3(a), confirming stable 8-channel WDM operation.

Experimental results

Figures 4(a) and (b) depict the measured 200-Gbaud PAM4 and PAM8 eye diagrams under back-to-back (BTB) and 2-km HCF transmission conditions. In the BTB configuration, the multi-level eye openings of the 200-Gbaud PAM signals are clearly distinguishable, benefiting from optimized transmitter pre-emphasis. After HCF transmission, moderate eye closure is observed, which is attributed to the accumulated transmitter, receiver, and fiber impairments.

Figure 4(c) shows the BTB BER curves of 200-Gbaud PAM4 and PAM8 signals, respectively. For PAM4, a BER of around 4.5×10^{-3} is achieved

at ROP of around 3 dBm, corresponding to the hard-decision forward error correction (HD-FEC) limit. For PAM8, a BER of around 5×10^{-2} is achieved at 6 dBm, corresponding to the soft-decision FEC (SD-FEC) threshold [15]. BTB BER performances of PAM4 and PAM8 signals at 140, 160, 180, and 200 Gbaud are depicted in Figure 4(d). With the ROP adjusted accordingly, the PAM4 BER increases from approximately 1×10^{-7} to 2×10^{-3} , while the PAM8 BER increases from approximately 4×10^{-3} to 5×10^{-2} as the baud rate increases from 140 to 200 Gbaud. The larger PAM8 penalty is mainly due to its reduced SNR margin.

Figure 4(e) shows the transmission penalties of 200-Gbaud PAM4 and PAM8 after 2-km HCF transmission. No significant additional BER penalty is observed, owing to the low chromatic dispersion near the operating wavelength.

Conclusion

In conclusion, we demonstrate an 8-channel cascaded O-band MRM transmitter with ~ 90 -GHz 3-dB electro-optic bandwidth, enabled by optimized carrier-depletion PN junctions and low-parasitic electrodes. The transmitter achieves 200-Gbaud PAM8 modulation with a record 600 Gb/s/ λ and 4.8-Tb/s aggregate capacity.

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